



NATIONAL DEVELOPMENT COMPLEX

Deputy Directorate General of Quality Control

Physical Laboratory: Site-3A

TEST REPORT

Test Report No: TR (016)/PL/2022

Job Card No: JC (030)/PL/2022

Date: 07-03-2022

INDENTER'S DATA

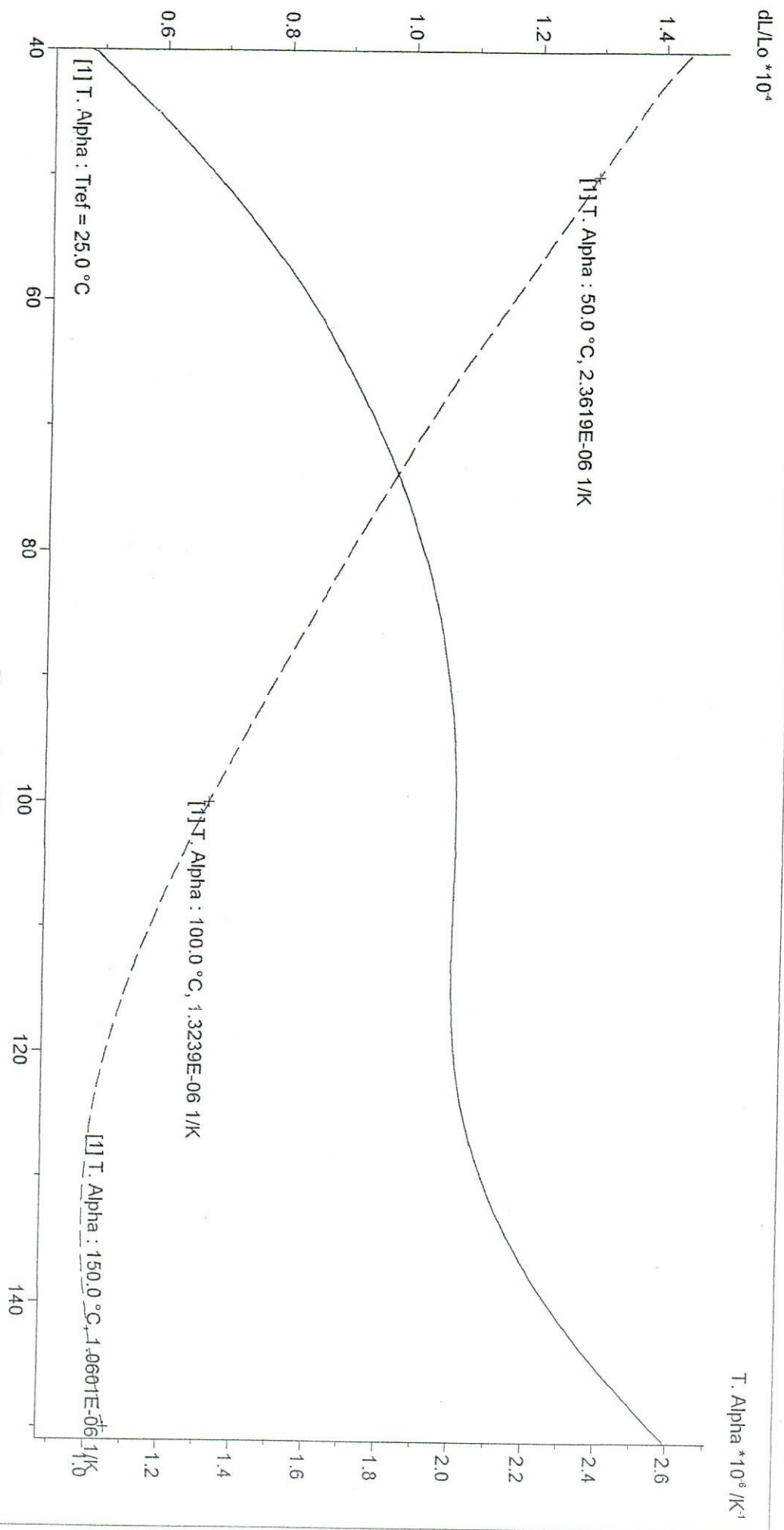
Name:	Shamim Khan Asif	Designation:	FM-1
Group & Site:	PP / 3C	Directorate/ Department:	ADP
Organization:	NDC	Contact No:	3656

GENERAL DATA

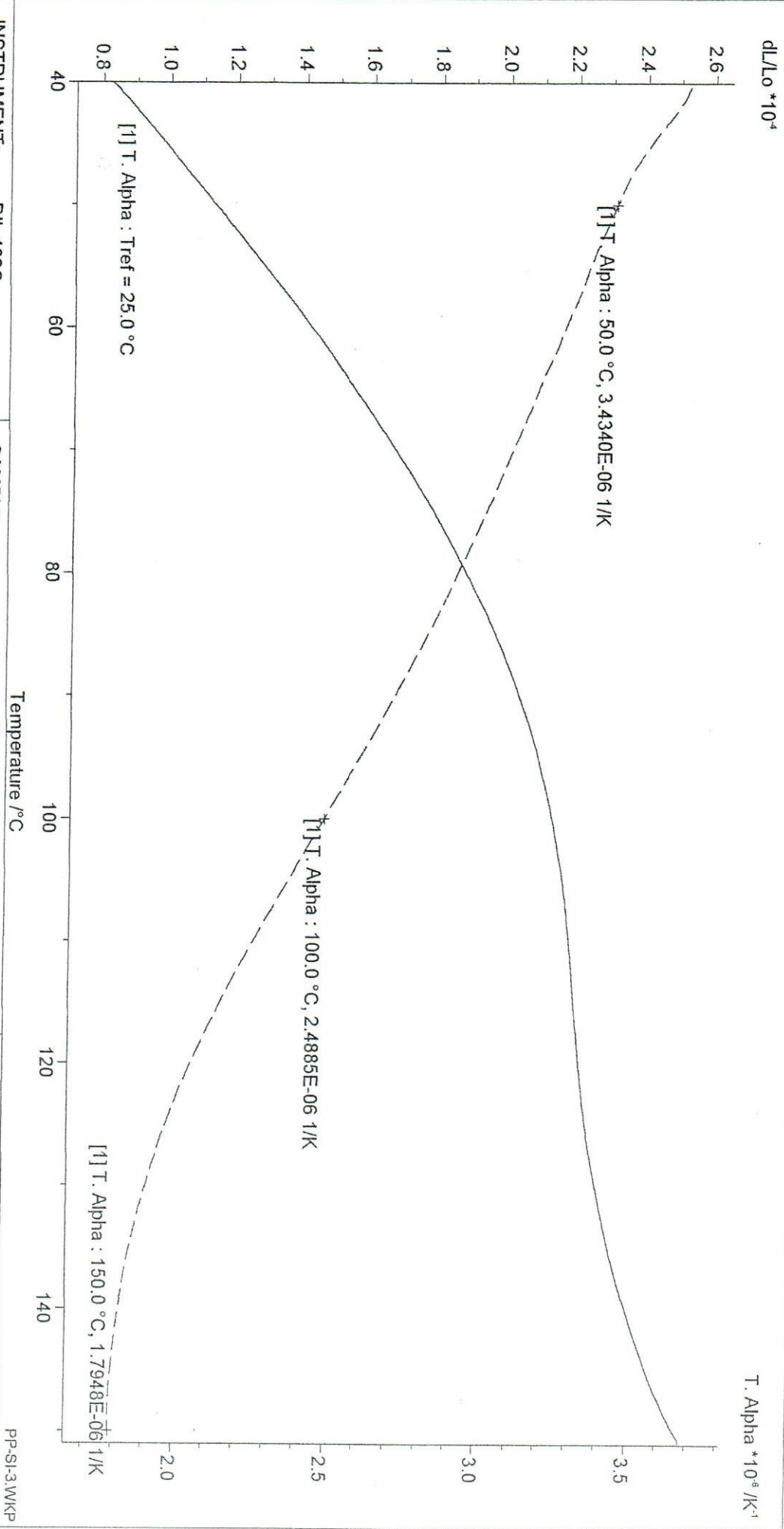
Sample Name:	Silica Phenolic Ablative	ID No:	PP-Si-10A/1120-LE-01~05
Lot No :	Si(820)-10A/1120-221~333	Lot Size:	7922RM
Shape:	Cylindrical	Chemical Composition:	Silica Phenol
Project:	All Systems	Quantity:	05
Sample Receiving Date:	07-03-2022	Test Date:	07-03-2022

TEST DATA

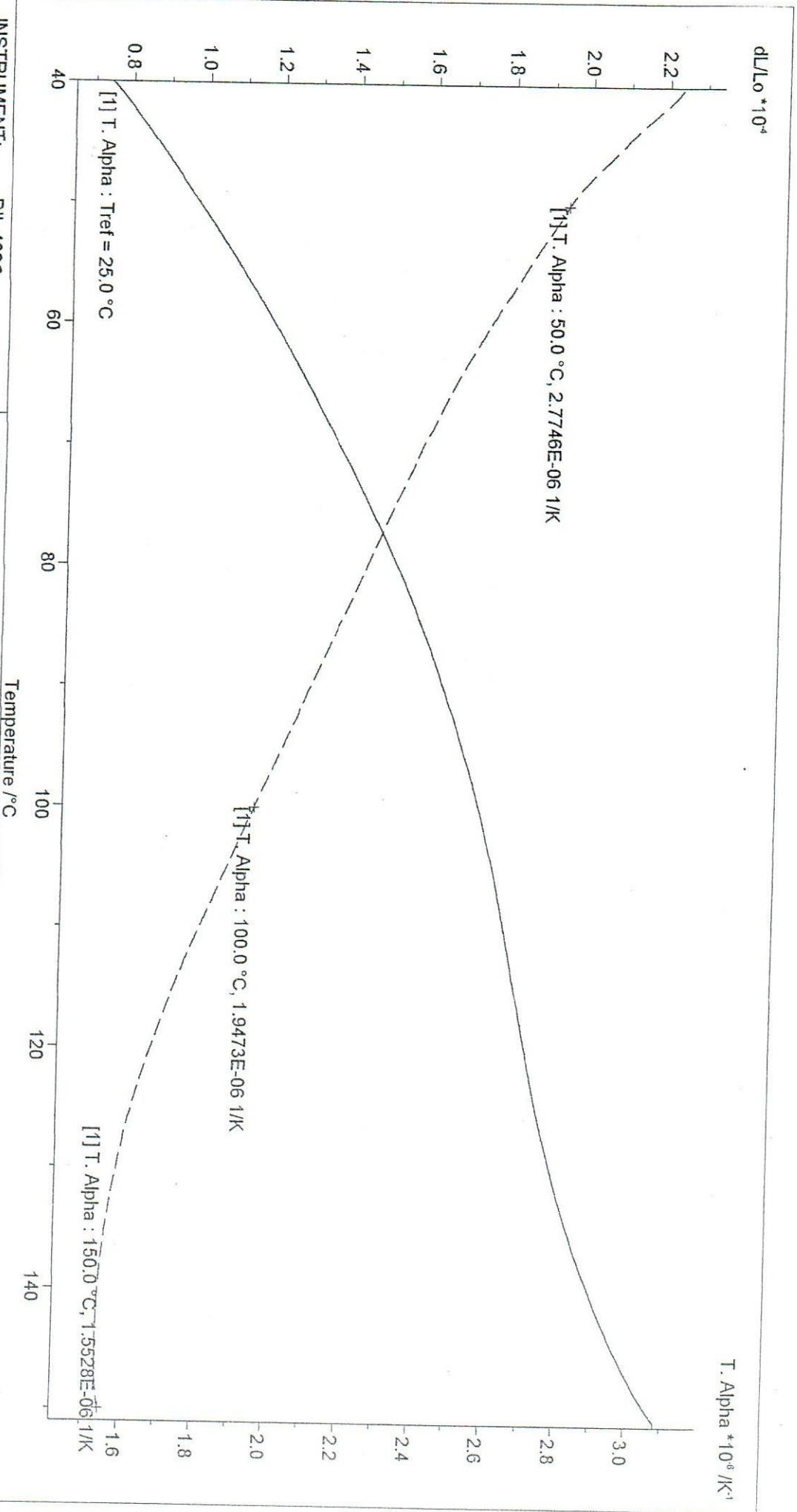
Test Method Used	PL/SOP/07	Qualification Criteria	NDC-Ab/QC/Si-C Ablatives/02, Rev : 01
Instrument Type	DIL-402C	Testing Temperature(°C)	RT~150
Temperature(°C)	21.7 ±1	Relative Humidity (%)	57 ±2
Test Type	Sample ID	Test Results	
Coefficient of Linear Thermal Expansion	PP-Si-10A/1120-LE-01	2.3619 E-06 1/K at 50°C	
		1.3239 E-06 1/K at 100°C 1.5819	
		1.0601 E-06 1/K at 150°C	
	PP-Si-10A/1120-LE-02	2.6850 E-06 1/K at 50°C	
		1.7524 E-06 1/K at 100°C 1.9559	
		1.4305 E-06 1/K at 150°C	
	PP-Si-10A/1120-LE-03	3.4340 E-06 1/K at 50°C	
		2.4885 E-06 1/K at 100°C 2.5724	
		1.7948 E-06 1/K at 150°C	
	PP-Si-10A/1120-LE-04	2.8311 E-06 1/K at 50°C	
		1.8361 E-06 1/K at 100°C 2.0185	
		1.3885 E-06 1/K at 150°C	



INSTRUMENT:	DIL 402C	SAMPLE:	Silica Phenolic Ablative / 24.410 mm	SAMPLE HOLDER TABLE:	AL203.SCL
FILE:	PP-SI-1.DLE	MATERIAL:	Silica Phenolic Ablative	SAMPLE HOLDER MATERIAL:	AI203
PROJECT:	MA	ATMOSPHERE:	N2/70	CALIBRATION FILE:	BL010222.CLE
IDENTITY:	PP-SI-10A/1120-LE-01	TEMP.CAL.FILE:	TCALNULL.TMX	CALIBRATION TABLE:	AL203.SCL
DATE/TIME:	07.03.22 10:06	RANGE:	25 $^{\circ}C$ /10.0(K/min)/1350 $^{\circ}C$	CALIBRATION MATERIAL:	AI203
LABORATORY:	Physical Lab, 3A	SEGMENTS:	1 / 1	CORRM.RANGE:	810/5000
OPERATOR:	Imran	MODE/TYPE OF MEAS.:	Standard expansion/samp.+corr.	REMARK:	NIL



INSTRUMENT:		DIL 402C	
FILE:		PP-SI-3.DLE	
PROJECT:		MA	
IDENTITY:		PP-SI-10A/1120-LE-03	
DATE/TIME:		07.03.22 12:47	
LABORATORY:		Physical Lab, 3A	
OPERATOR:		Imran	
SAMPLE:		Silica Phenolic Ablative / 24.380 mm	
MATERIAL:		Silica Phenolic Ablative	
ATMOSPHERE:		N2/70	
TEMP.CAL.FILE:		TCALNULL.TMX	
RANGE:		25 $^{\circ}C$ /10.0(K/min)/1350 $^{\circ}C$	
SEGMENTS:		1 / 1	
MODE/TYPE OF MEAS.:		Standard expansion/samp.+corr.	
SAMPLE HOLDER TABLE:		AL2O3.SCL	
SAMPLE HOLDER MATERIAL:		AL2O3	
CALIBRATION FILE:		BL010222.CLE	
CALIBRATION TABLE:		AL2O3.SCL	
CALIBRATION MATERIAL:		AL2O3	
CORRM.RANGE:		810/5000	
REMARK:		NIL	



INSTRUMENT:	DIL 402C	SAMPLE:	Silica Phenolic Ablative / 24.490 mm
FILE:	PP-SI-5.DLE	MATERIAL:	Silica Phenolic Ablative
PROJECT:	MA	ATMOSPHERE:	N2/70
IDENTITY:	PP-SI-10A/1120-LE-05	TEMP.CAL.FILE:	TCALNULL.TMX
DATE/TIME:	07.03.22 15:22	RANGE:	25 °C/10.0(K/min)/1350 °C
LABORATORY:	Physical Lab, 3A	SEGMENTS:	1/1
OPERATOR:	Imran	MODE/TYPER OF MEAS.:	Standard expansion/samp.+corr.
		SAMPLE HOLDER TABLE:	AL203.SCL
		SAMPLE HOLDER MATERIAL:	AI203
		CALIBRATION FILE:	BL010222.CLE
		CALIBRATION TABLE:	AL203.SCL
		CORR/MATERIAL:	AI203
		REMARK:	810/5000
			NIL